

Assessing Bicycle Lane Quality Using Smartphone Sensor Data

5ARE0 - Assignment 1 - Data collection and analysis report (CRISP-DM)

Group X - members and student numbers: [fill in: names + student numbers of all group members]

1. Business / Objective Understanding

Problem. Bike-lane surface defects (cracks, potholes, uneven patches) make cycling uncomfortable and raise the risk of accidents, but inspecting lanes by hand is slow and expensive. Smartphones already carry accelerometers, gyroscopes and gravity sensors, so every cyclist could act as a mobile road-quality sensor. **Aim:** build and evaluate a pipeline that labels a stretch of bike lane as *smooth* or *bumpy* from the motion sensors of a phone carried in a pocket, and apply it to a new, independent dataset to map which road sections are smooth and which are bumpy.

Relevance to health monitoring. Vibration and shocks transmitted to the rider are a comfort and safety factor, and a rider-worn sensor that scores surface roughness per second of riding is a natural building block for vibration-exposure monitoring and for prioritising repairs where riders are most affected. In this study we only establish the technical building block (surface roughness classification); links to actual health outcomes were not measured and are not claimed.

Research questions. (RQ1) Can smooth and bumpy lanes be told apart from phone motion sensors? (RQ2) Which features and which of four supervised / four unsupervised models work best, and why? (RQ3) Does the result transfer to a rider the model has never seen? RQ3 sets our success criterion: performance is judged on **held-out participants**, not on random windows.

2. Data Understanding

2.1 Collection methodology

Three group members (P1-P3; anonymized) each recorded one smooth and one bumpy ride with the Sensor Logger app: accelerometer, gyroscope and gravity sensor at 100 Hz, all other sensors off, *Standardisation* on. Following the protocol, the phone was put in the pocket, the rider rode a continuous stretch of lane, then stopped the recording. *Smooth* = a typical red cycling lane without imperfections; *bumpy* = a commonly used lane with noticeable, non-dangerous imperfections. Recording length was left to the group; it was set by the stretch of lane chosen (48-197 s), and modeling uses short windows, so length only changes how many windows a ride contributes. Phone / placement: P1 iPhone 15 (pocket); [fill in: phone models, exact placement (pocket, side, orientation) and bike type for P2 and P3, and route/location of each ride].

Ride	Duration (s)	Samples / sensor	Recorded (UTC)	Eff. rate (Hz)	Mean phone tilt
P1 smooth	77.3	7 719	2026-09-28	99.9	1.9°
P1 bumpy	180.3	18 010	2026-09-13	99.9	4.9°
P2 smooth	126.7	12 576	2026-09-28	99.2	7.1°
P2 bumpy	197.1	19 553	2026-09-12	99.2	19.9°
P3 smooth	47.8	4 763	2026-09-10	99.6	48.6°
P3 bumpy	76.1	7 583	2026-09-12	99.6	22.6°

Table 1. Recordings used (11.8 min in total). Tilt = angle between the mean gravity vector and the phone y-axis.

2.2 Data quality and label quality

Technical quality is good: no missing values, no duplicated timestamps, effective sampling 99.2-99.9 Hz, largest gap 16 ms, and the three sensors share one clock (row-for-row identical timestamps), so no resampling or interpolation was needed. **Label quality was the real problem.** A first round of "smooth" rides by P1 and P2 turned out to include lane imperfections: their spread and peaks were far above genuinely smooth rides (std of $|a|$: P1 1.55 vs 0.97 after re-recording, peak 26.6 vs 10.2 m/s²; P2 2.41 vs 0.93). Both rides were rejected *before* any model comparison, re-recorded on cleaner stretches, and are kept separately (*data/raw_excluded*) for transparency. The final smooth rides of all three participants agree (std 0.76-0.97), while all bumpy rides are much higher (2.67-3.47).

2.3 Exploratory data analysis

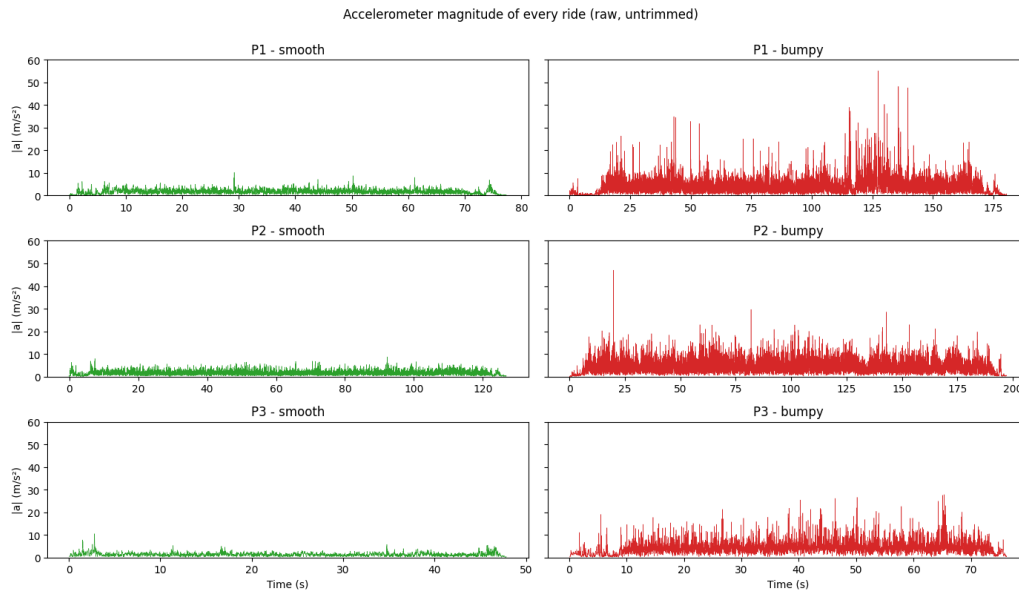


Figure 1. Accelerometer magnitude of every ride (raw). Within each participant the bumpy ride has a much heavier tail. Note the quiet first seconds of the bumpy rides of P1 (~12 s) and P3 (~8 s).

- **Separation holds within and across people:** for every participant bumpy windows show higher spread and peaks, and the same ordering repeats for all three riders - the precondition for transferring to an unseen rider (RQ3).
- **The difference is broadband:** the mean power spectrum of $|a|$ is about ten times higher for bumpy lanes across the whole 0.5-50 Hz range, not in one narrow band - mostly a difference in vibration intensity.
- **Whole-ride labels are noisy at the edges:** the rider is not yet on the rough section / still pulling away in the first seconds, yet the ride is labeled bumpy. This foreshadows the error analysis in Section 5.5.
- **Phone tilt differed between rides** (2°-49°, Table 1; P3's smooth ride was carried at about 49°) - a placement inconsistency with consequences for feature choice (Section 3.3).

3. Data Preparation

3.1 Preprocessing and windowing

Merge: accelerometer, gravity and gyroscope files are joined column-wise after asserting identical timestamps. **Trim:** the first and last 1.5 s of each ride are removed (phone handling, pulling away, coming to rest). Two *adaptive* trims were tried first and rejected on all six rides: a fixed rolling-std threshold did not transfer between phones/riders (baseline vibration differs), and a per-ride percentile threshold silently deleted up to 43 s of real riding. A small fixed, *bounded* trim cannot destroy data and is therefore used. **Windows:** 2 s (200 samples) with 50 % overlap, causal (only past samples), giving 675 windows (237 smooth, 438 bumpy). Window length was tested rather than assumed (leave-one-participant-out macro-F1, Logistic Regression): 1 s 0.947/0.942 (0 %/50 % overlap) - clearly worse; 2 s 0.953/0.960; 3 s 0.970/0.967; 4 s 0.971/0.971. Beyond 2 s the gain is about 0.01, within noise for three folds, while longer windows cost spatial resolution (2 s is about 10 m of lane at 5 m/s) and real-time latency, so 2 s / 50 % was kept.

3.2 Feature engineering

For each window six statistics (mean, std, min, max, RMS, IQR) were computed on 11 channels (accelerometer x/y/z and magnitude, gyroscope x/y/z and magnitude, gravity x/y/z): 66 candidate features. A frequency-domain set (dominant frequency, four band energies, spectral entropy, zero-crossing rate on both magnitudes; 14 features) was also tested.

3.3 Feature selection strategy

Orientation-invariant features. Gravity features describe how the phone is tilted, not how rough the road is, and per-axis features depend on how the phone sits. The evidence: P3's smooth ride (tilt 49°) made gravity features about ten times more different from the other smooth rides than accelerometer/gyro features were, and DBSCAN kept isolating those windows as their own cluster. Three feature sets were compared: A all 66, B without gravity (48), C magnitude-only (12: six statistics of |accelerometer| and |gyroscope|). C is the only set independent of phone orientation and was best or tied best for every model (GMM ARI: A 0.465, B 0.433, C 0.775; K-Means 0.675 / 0.680 / 0.715; Agglomerative 0.856 for all). This also matches deployment, where placement is not controlled. **Frequency features did not help:** on their own macro-F1 was only 0.69-0.80, and adding them to the time-domain features left performance flat or lower (Logistic Regression 0.960 -> 0.913). **Selection:** ANOVA F, mutual information and Random-Forest importance were combined (Section 5.3), and the number of selected features k was tuned inside the nested cross-validation.

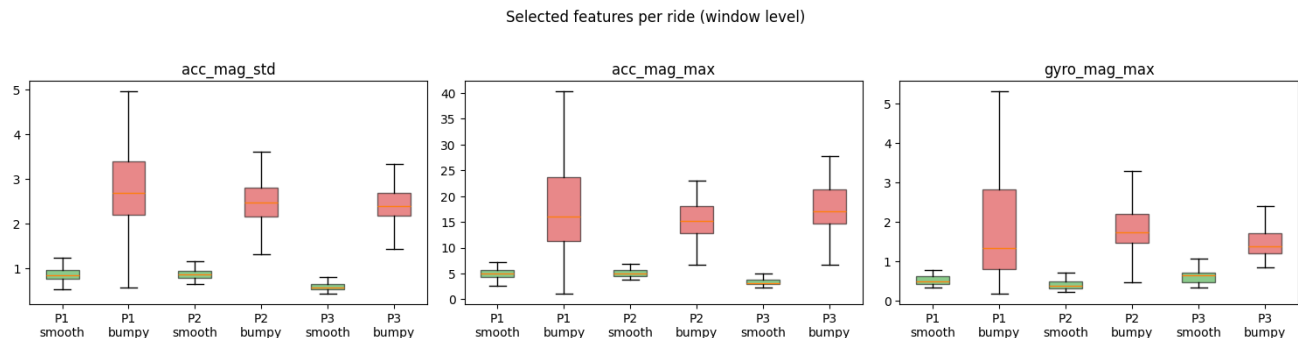


Figure 2. Three of the selected features per ride (window level). Bumpy rides are clearly higher for all three participants.

4. Modeling

4.1 Unsupervised approaches (labels never used for fitting)

Features (set C) were standardized and reduced with PCA to 4 components (95 % of the variance). Four models with different assumptions were fitted: **K-Means** ($k = 2$, partitional), **Agglomerative clustering** (Ward linkage, $k = 2$, hierarchical), **DBSCAN** (density-based, $\text{min_samples} = 5$) and a **Gaussian Mixture Model** (2 components, probabilistic). DBSCAN is very sensitive to eps : the k -distance percentile rule ($\text{eps} = 1.31$) merged nearly everything into one cluster ($\text{ARI} \sim 0$), whereas $\text{eps} = 0.3$ - 0.6 separates the classes; $\text{eps} = 0.6$ was used. Since that choice was made while looking at label-based scores, DBSCAN's result is optimistic ("what it can do when well tuned").

4.2 Supervised approaches

Four classifiers from four model families, in a pipeline *StandardScaler* -> *SelectKBest(k)* -> *classifier* (scaling and selection are refit on training data only): **Logistic Regression** (C in $\{0.1, 1, 10\}$), **Decision Tree** (max depth $\{2, 3, 5, 8\}$, min leaf $\{5, 15\}$), **Random Forest** (trees $\{100, 300\}$, depth $\{3, 6, \text{none}\}$) and **SVM with RBF kernel** (C $\{0.1, 1, 10\}$, γ $\{\text{scale}, 0.1\}$; k in $\{4, 6, 12\}$ for all. Grids are deliberately small for ~ 675 windows. Class imbalance (438 vs 237) is handled with balanced class weights, and results are reported with balanced accuracy and macro-F1 as well as accuracy. **Validation is nested leave-one-participant-out (LOPO):** train on two riders, test on the third, repeat; hyperparameters are tuned only on the two training riders (inner grouped 2-fold CV, macro-F1). This is much stricter than a random split, because overlapping windows of one ride are near-duplicates and would leak.

4.3 Comparison methodology

Supervised models are scored on unseen participants (accuracy, balanced accuracy, macro-F1, confusion matrices). Unsupervised models are scored internally (silhouette) and externally against the true labels (Adjusted Rand Index, purity). **The two scores are not equivalent:** unsupervised numbers are in-sample (clusters are fitted on all windows), supervised numbers are out-of-participant. A model is chosen by a rule fixed beforehand: best mean macro-F1, and among models within 0.01 of it the simplest and most interpretable.

5. Evaluation

5.1 Unsupervised results

Model	Clusters	Noise	Silhouette	ARI	Purity
K-Means	2	0 %	0.477	0.715	0.923
Agglomerative (Ward)	2	0 %	0.472	0.856	0.963
DBSCAN (eps 0.6)	6	32.7 %	0.262	0.442	0.950
GMM	2	0 %	0.446	0.775	0.941

Table 2. Unsupervised models (in-sample). ARI and purity compare clusters with the true smooth/bumpy labels.

Agglomerative clustering recovers the classes best: 234 of 237 smooth windows fall in one cluster and 416 of 438 bumpy windows in the other. The two main clusters of K-Means, Agglomerative and GMM are each spread over all three participants, so they describe road roughness and not rider identity - which was not the case before the data and feature fixes (first run: Agglomerative ARI 0.585, DBSCAN 0.041, GMM 0.042). DBSCAN forms a clean smooth core (211 smooth vs 8 bumpy) but labels 195 of 438 bumpy windows as noise: smooth riding is uniform and dense, bumpy riding is varied.

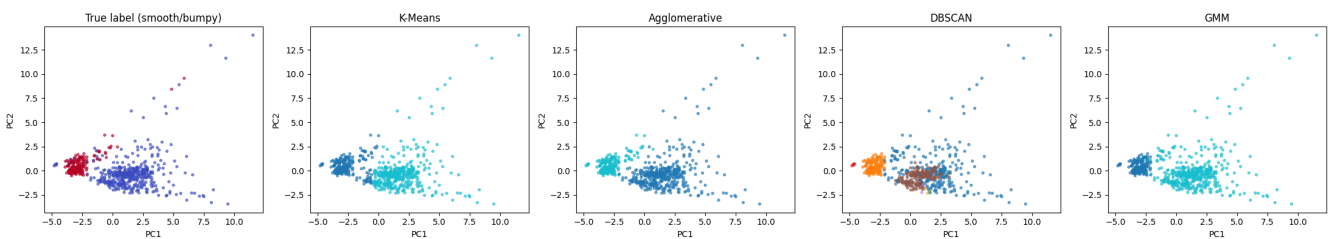


Figure 3. PCA projection: true labels and the four clusterings.

5.2 Supervised results (leave-one-participant-out)

Model	Accuracy	Balanced acc.	Macro-F1 (mean $\pm$ sd)	F1 on P1	F1 on P2	F1 on P3
Logistic Regression	0.960	0.964	0.955 $\pm$ 0.034	0.916	0.977	0.972
Decision Tree	0.931	0.929	0.924 $\pm$ 0.044	0.912	0.973	0.887
Random Forest	0.952	0.954	0.947 $\pm$ 0.018	0.930	0.966	0.944
SVM (RBF)	0.961	0.968	0.957 $\pm$ 0.031	0.921	0.977	0.972

Table 3. Supervised models, each tested on a participant never seen in training (mean of three folds).

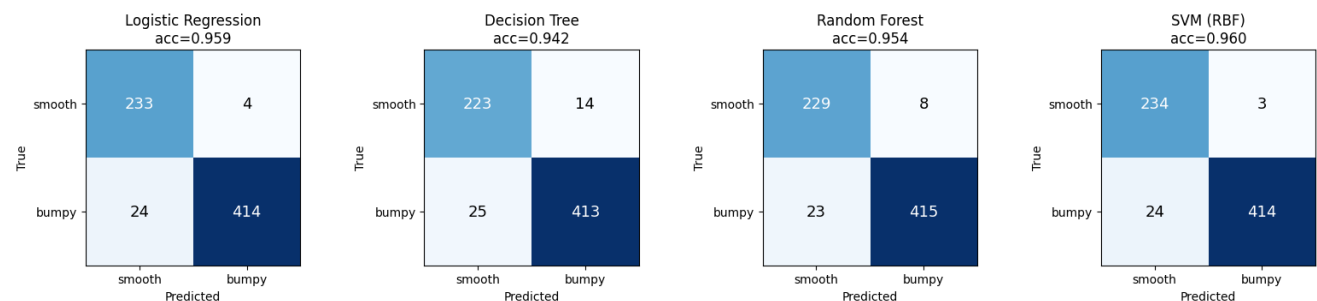


Figure 4. Pooled out-of-fold confusion matrices (every window predicted by a model that never saw its rider; bumpy = positive class).

Logistic Regression, SVM and Random Forest are statistically tied (differences of 0.01 are within noise with three folds); the Decision Tree is clearly weakest and least stable (F1 0.887 on P3). P1 is the hardest rider for every model (F1 0.91-0.93), which is explained in Section 5.5.

5.3 Feature selection and optimal features

The accelerometer-magnitude statistics rank highest under all three methods: RMS (mean rank 2.3), std (2.7), mean (3.0), IQR (3.3) and max (4.0), followed by gyroscope max (5.7). Under nested LOPO, **a single feature already reaches macro-F1 0.946 and 4-6 features reach the plateau (~0.96)**; the gyroscope adds little. The selected model uses $k = 6$: acc_mag mean, std, max, RMS, IQR and gyro_mag max. Physically this fits: rough lanes produce stronger shocks, so vibration intensity is what separates the classes.

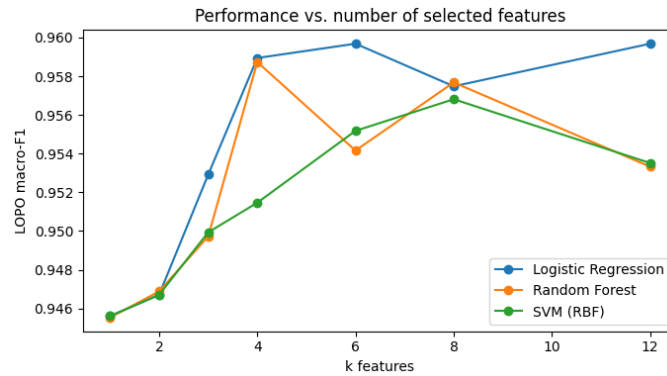


Figure 5. LOPO macro-F1 versus number of selected features (selection inside each training fold).

5.4 Model choice and justification

Model	Macro-F1	Inference (μs/window)	Tuning time (s)	Features (k)	Interpretability
Logistic Regression	0.955	0.5	0.2	6	High: one signed weight per feature
Decision Tree	0.924	0.5	0.4	12	High: readable if-then rules
SVM (RBF)	0.957	2.3	0.3	12	Low: kernel, no direct weights
Random Forest	0.947	9.5	13.2	12	Low-medium: importances only

Table 4. Comparison on the five assignment criteria (timings from one run on the analysis machine, single core).

Selected model: Logistic Regression (three models within 0.01 F1 of the best; it is the simplest). *Problem suitability:* the classes separate mainly along one axis (vibration intensity), so a linear boundary suffices; RF and SVM gain nothing. *Data requirements:* ~675 windows from three riders favours a low-capacity, regularized model; the high-variance tree was weakest. *Interpretability:* one signed weight per feature (largest: acc_mag std, max, IQR), explainable to a city planner; correlated features share weight, so weights are read as a group. *Efficiency:* under a microsecond per window and no heavy state - feasible on a phone in real time. *Performance:* tied with the best model on unseen riders.

5.5 Supervised vs. unsupervised, and error analysis

Effectiveness. The best unsupervised model (purity 0.96, ARI 0.86) and the supervised models (accuracy 0.96, macro-F1 0.95) look similar, but the supervised numbers come from the stricter test on unseen riders, so they are the stronger evidence. That clustering independently recovers the same structure shows the labels reflect a real signal. **Advantages/limits.** Unsupervised methods need no labels (useful for surveying new cities), can reveal unlabeled surface types and can expose label problems; but they cannot name clusters, need a hand-chosen k or eps, and DBSCAN was fragile. Supervised models give a direct smooth/bumpy verdict with probabilities and can be validated honestly, but need labeled rides and inherit label noise. **Preferred when:** supervised for deployment (a verdict per road section); unsupervised for exploration, label sanity checks and discovering new surface types.

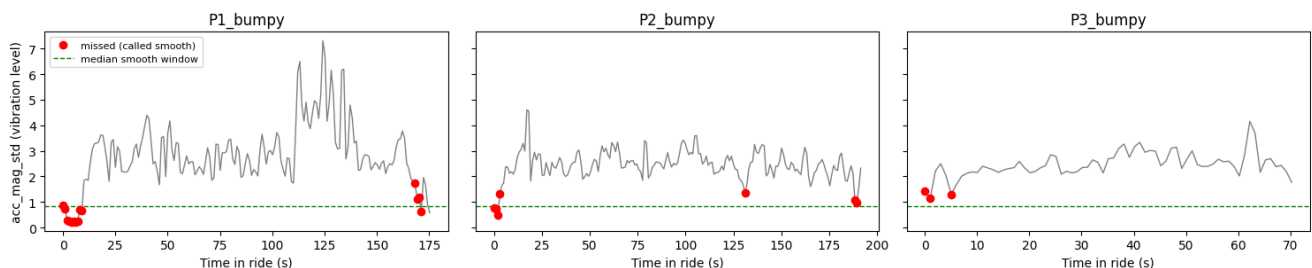


Figure 6. Vibration level over time for each bumpy ride; red dots = windows the selected model called smooth.

Error analysis. The selected model makes 28 errors in 675 windows; **27 fall in the first or last 10 s of a ride** (17 of 30 bumpy windows missed in the first 10 s, 6 of 30 in the last 10 s), while the middle of rides has 1 error in 555 windows (0.2 %). The missed windows have a vibration level (median std 0.75) almost equal to smooth windows (0.82), so the signal itself looks smooth. Two explanations: (i) **label noise** - whole rides are labeled bumpy, including their quiet start and end (Figure 1); (ii) a plausible **speed effect** - vibration grows with speed, so pulling away or slowing down makes even a bumpy lane feel gentle (speed was not recorded, so this is untested). Smooth windows are almost never called bumpy (4 of 237). The edge trim was not lengthened afterwards to improve scores, because that choice would be made after seeing the errors.

6. Deployment Considerations & Recommendations

6.1 Deployment pipeline and demonstration

The deployed pipeline is: load the three sensor files -> (files only) drop 1.5 s at each end -> causal 2 s windows every 1 s -> six magnitude features per signal -> scaler, selector and Logistic Regression -> smooth/bumpy and P(bumpy) per window. It only needs the past 2 s, so it works on a live stream, and the whole model is 6 weights plus an intercept and the scaler. For plots, predictions can be smoothed with a causal majority vote over three windows. A route map is drawn when GPS (Location.csv) is present; otherwise a timeline is shown. **Demonstration without external data:** with the model trained *without* P3, the P3 smooth ride is classified 100 % correctly (43 windows) and the P3 bumpy ride 95.8 % correctly (71 windows; Figure 7).

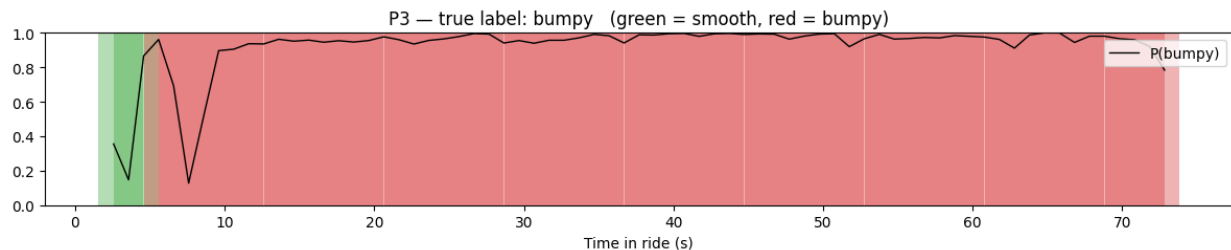


Figure 7. Held-out rider P3, bumpy ride: predicted class per window (red = bumpy) and P(bumpy). The first seconds are called smooth (see error analysis).

External dataset. The independent dataset was not used for training, tuning or testing. [after applying the model: report the share of windows predicted smooth/bumpy, insert the route or timeline figure, and discuss whether it matches what is known about the route]

6.2 Limitations

- **Small study:** 3 riders, 6 rides; LOPO has only 3 folds, so model differences of ~ 0.01 are noise, and each rider rode a single stretch per class - lane and label are entangled, so the model may partly learn those roads.
- **Placement and hardware:** tilt ranged from 2° to 49° across rides; magnitude features are robust to tilt but not to a different mounting (pocket vs handlebar changes vibration transmission), phones or bikes.
- **Speed not recorded:** the largest failure mode (ride edges) is probably speed related. **Label granularity:** one label per ride. **Selection optimism:** feature-set choice and DBSCAN eps were made while viewing label-based scores.
- **Preprocessing:** the 1.5 s trim applies to recorded files only; a live stream has no start or end. Frequency-domain features may become useful with more riders and bikes.

6.3 Recommendations and future work

- Collect more riders, bikes, phones and routes, several rides per class, with a written placement standard (same pocket, orientation, tight fit) and GPS speed; annotate quiet approach segments (Sensor Logger annotations) so labels are per segment, not per ride.
- At deployment, use speed to flag or skip low-speed segments (start-ups, junctions), smooth predictions over consecutive windows, use windows of fixed road length (GPS distance) instead of fixed time, and re-check accuracy on each new phone/placement.
- **Practical applications:** city maintenance prioritisation from crowd-sourced rides; rider-facing comfort/route advice. **Health-related use:** a per-second roughness score could serve as a proxy for vibration exposure in cyclist comfort or injury-risk studies - but this would require validation against measured exposure or clinical outcomes, which this study did not do.

6.4 Ethics, privacy and reproducibility

All participants are consenting adult group members who understood the purpose; safety came first (no hazardous lanes); data are anonymized (P1-P3; device identifiers from the app metadata were not shared), used only for this assignment and will be deleted after the course. The submitted notebook runs end-to-end from *data/* using only numpy, pandas, matplotlib and scikit-learn (versions listed at its top and in the README); raw data, processed data and rejected recordings are stored separately.